



Robust and Multi-Objective Alignment of Large Language Models via Reinforcement Learning from Human Feedback: A Statistical Perspective

Huyen-Trang Le, Computer Network Master Program – M2 CNS SR

Under the supervision of Dr. Pegah Alizadeh and Prof. Stéphane Chrétien

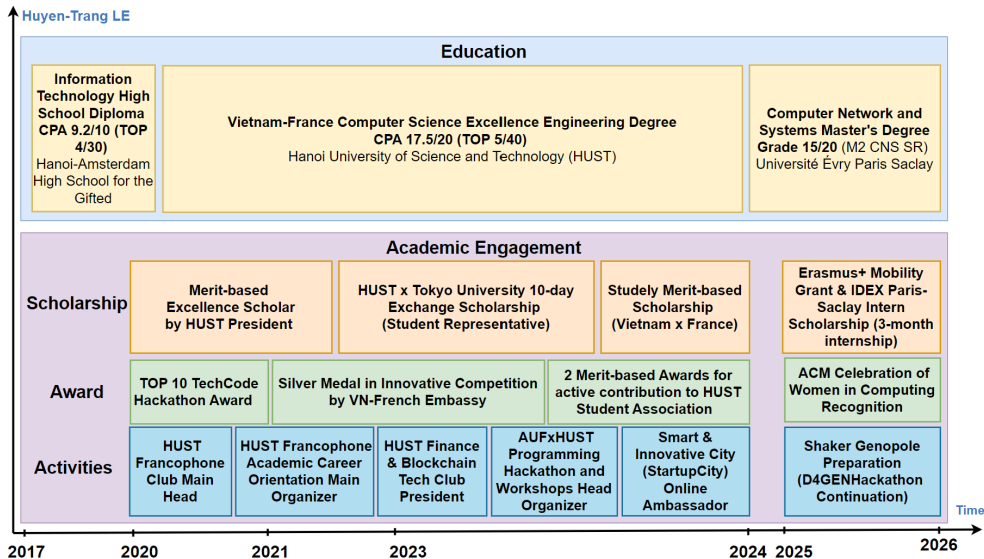
Doctor of Philosophy Research Proposal, ERIC Laboratory, Université Lumière Lyon 2

Main Content

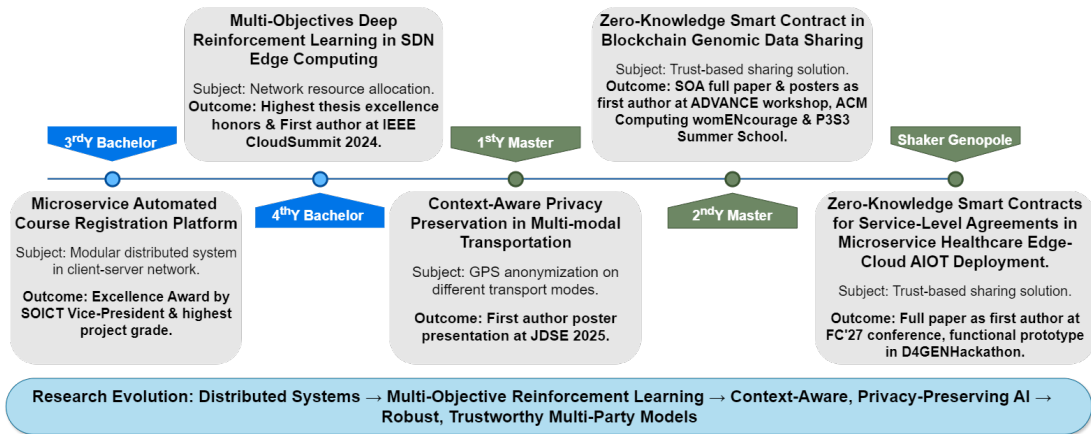
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1.1 Personal Curriculum Vitae: Education & Academic Engagement



1.2 Personal Curriculum Vitae: Research Experience



2.1 PhD Thesis Description: Research Context and Literature Review

Aligning Large Language Models with Human Feedback

Large Language Models (LLMs) pretrained on massive corpora do not automatically behave in ways that are **helpful, harmless, and honest** [1]. **Reinforcement Learning from Human Feedback (RLHF)** fine-tunes LLMs using **human preference signals** rather than hand-crafted rewards [2].

- **Proximal Policy Optimization (PPO)** [3] is the classical RL engine behind RLHF: it trains a policy against a learned **reward model** using a **clipped surrogate objective** for stable updates.
- **Direct Preference Optimization (DPO)** [4] reframes alignment as a **closed-form classification loss** on preference pairs, removing the explicit reward model and the RL loop entirely.
- **Group Relative Policy Optimization (GRPO)** [5] drops the learned value network, estimating a **baseline from a group of sampled outputs**, reducing memory and improving efficiency $\implies$ **Key Challenges** remain open.



2.1 PhD Thesis Description: Research Context and Literature Review

3 Key Challenges (KC) in RLHF Alignment of LLMs

Robustness to Imperfect Feedback (KC1)

Policies trained with **PPO**, **DPO**, or **GRPO** can **overfit noisy, biased, or adversarial preference data**, exploiting blind spots of the reward signal (**reward hacking**) and degrading real-world reliability.

Single-Objective Bias (KC2)

Human values are **multiple and often competing**, yet most **PPO/DPO/GRPO** pipelines optimize a **single scalarized reward**.

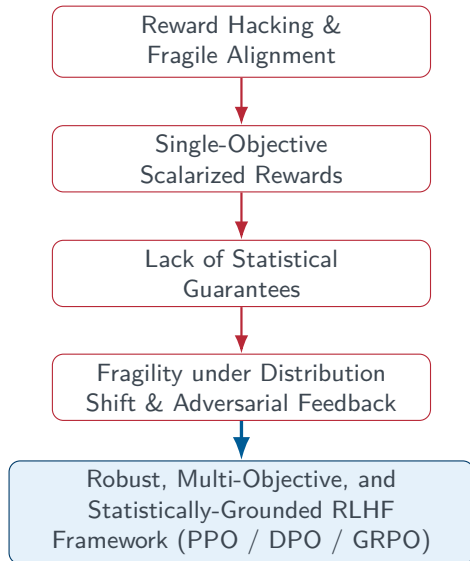
Missing Statistical Guarantees (KC3)

Sample complexity, convergence, and uncertainty quantification of preference-based RL fine-tuning remain **poorly understood theoretically**.



2.1 PhD Thesis Description: Research Context and Literature Review

- **PPO-based RLHF** [3, 1] remains vulnerable to **reward hacking** and **training instability** at scale.
- **DPO** [4] improves stability and simplicity but is derived under **idealized, noise-free preference** assumptions.
- **GRPO** [5] improves efficiency for reasoning tasks but **inherits a single-reward formulation** and lacks **multi-objective** treatment.
- Most works optimize **one reward signal in isolation**, lacking **principled statistical analysis** or **Pareto-aware trade-offs** across competing human values.



2.2 PhD Thesis Description: Scientific Objectives (SO)

Propose a Robust, Multi-Objective, and Statistically-Grounded Reinforcement Learning Framework for Aligning Large Language Models with Human Feedback.

SO1: Robust Preference Optimization & Reward-Hacking Mitigation

- **Robustness (KC1):** Design RL/DPO/GRPO variants **resilient** to noisy, biased, or adversarial preference data.

SO2: Multi-Objective Alignment across Competing Human Values

- **Multi-Objective Alignment (KC2):** Extend PPO/DPO/GRPO to **jointly optimize** helpfulness, harmlessness, and honesty via **Pareto-aware or constrained** formulations.

SO3: Statistical Foundations for Preference Optimization

- **Statistical Guarantees (KC3):** Derive sample-complexity, convergence, and **uncertainty quantification** results for the PPO/DPO/GRPO family of algorithms.



2.3 PhD Thesis Description: Research Questions (RQ) and Methodological Approaches (MA)

RQ1: How to design preference-optimization algorithms robust to noisy or adversarial human feedback and resistant to reward hacking (SO1)?

- **Robust Reward Modeling (MA1):** Uncertainty-aware / ensemble reward models and explicit **preference-noise** modeling.
- **Robust Policy Optimization (MA2):** Regularized extensions of **PPO's clipped objective** and **GRPO's group-relative baseline**.

RQ2: How to formulate and solve multi-objective alignment as a principled extension of DPO/PPO/GRPO across competing objectives (SO2)?

- **Multi-Objective Preference Optimization (MA3):** Scalarization vs. **Pareto-front** / constrained optimization.
- **Group-Relative Multi-Reward Extensions (MA4):** Extending **GRPO's group baseline** to **multiple, conflicting** reward signals.



2.3 PhD Thesis Description: Research Questions (RQ) and Methodological Approaches (MA)

RQ3: How to establish statistical guarantees (sample complexity, convergence, uncertainty) for **preference-based RL alignment** methods **(SO3)**?

- **Statistical Learning Theory for Preference Optimization (MA5):** Theoretical analysis building on **DPO**'s reduction of RLHF to a classification problem.
- **Convergence & Uncertainty Analysis (MA6):** Finite-sample and asymptotic analysis of **PPO/GRPO-style** policy updates under preference data.



3. Working Method and Thesis Working Plan: 3-Year Workflow Plan

	Phase 1 SOA & Preference/Reward Modeling			Phase 2 Multi-Obj. Algorithm Design		Phase 3 Prototype & Stat. Analysis		Phase 4 Evaluation & Validation	
Task	10/2026	03/2027	06/2027	12/2027	03/2028	06/2028	03/2029	06/2029	10/2029
Literature Review & SOA	RLHF, PPO/DPO/GRPO, LLM Alignment								
Feedback-Noise & MO Value Model	Feedback noise & MO value model								
Algorithm Design & Theoretical Analysis		Robust Multi-Objective RLHF Framework Formulation							
System / Prototype Development				Algorithm Implementation & Training Pipeline (PPO/DPO/GRPO)					
Performance Evaluation & Validation						Benchmark vs existing RLHF baselines			
PhD Defense								Prepare thesis documents	
Summer Schools & Academic Activities				Summer Schools, Workshops & Winter Schools			Open-Source Robust / Multi-Objective RLHF Library		
Scientific Perspective		Publish IEEE/ACM ranked conference			Publish IEEE/ACM ranked conference			Publish Q1 journal	

Analyze
Design
Develop
Review/Test
Contribute



4. Conclusion: Motivation to Pursue a PhD

Pathway to Academic Career

My passion for science, perseverance, and strong motivation.

- **Inspiration:** from **Dr. Alizadeh** and **Prof. Chrétien**, whose work on **reinforcement learning, preference optimization, and statistical learning**.
- **Goal:** to become a **Researcher and Lecturer** contributing to **scientific progress, training and supporting students (in particular those facing challenges similar to mine as I was supported once by my professors)**, and promoting **Young Women in Science**.
- **Motivation:** **My late mother (university professor)** and **Vietnamese professors** who provided me **financial & educational support** for my graduate studies.



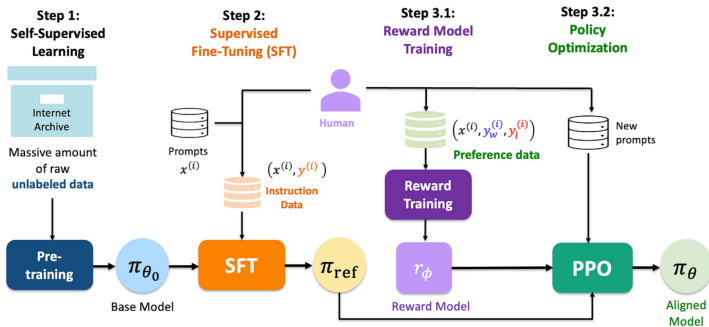
Thank you for your attention!

I am open to your questions

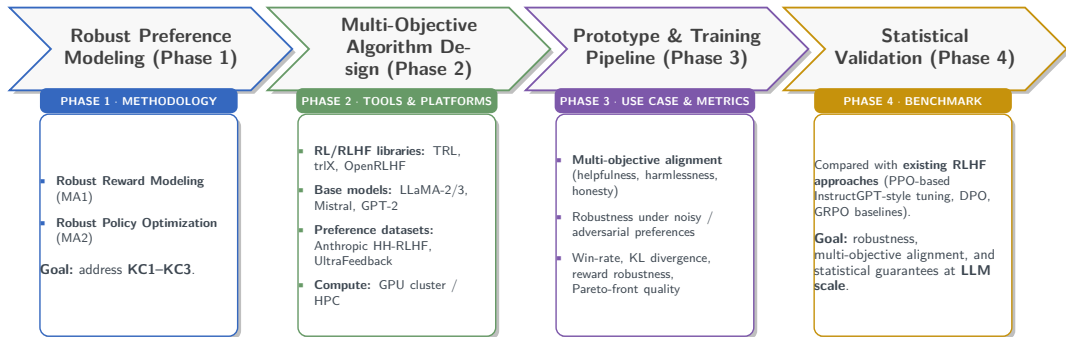
Conceptual Architecture

The proposed framework encompasses **four stages**: **human preference collection**, an **uncertainty-aware reward model**, **robust policy optimization** (PPO / DPO / GRPO variants), and a **multi-objective aggregation & statistical monitoring** layer that tracks convergence and uncertainty throughout training.

Reinforcement Learning from Human Feedback (RLHF)

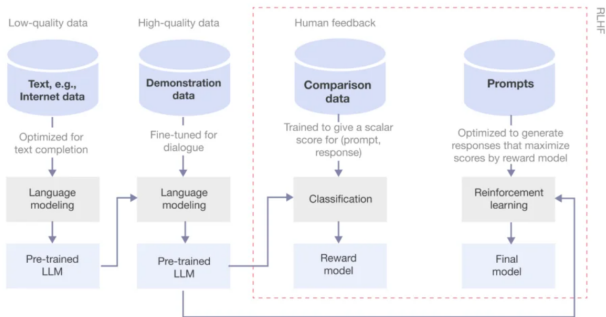


PhD Thesis Description: Working Method



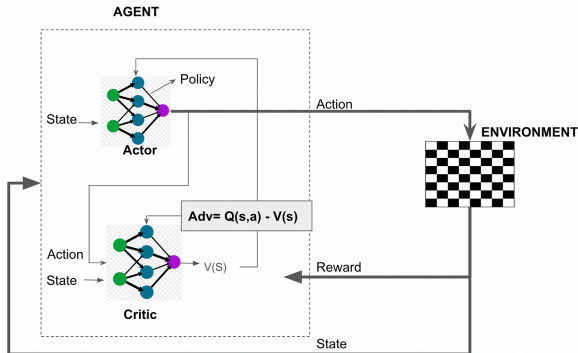
Reinforcement Learning from Human Feedback (RLHF)

A pipeline that fine-tunes an LLM using **human preference judgments** rather than a hand-crafted reward [2, 1]: **supervised fine-tuning** → **reward modeling** → **RL fine-tuning**.



Proximal Policy Optimization (PPO)

A **policy-gradient RL algorithm** [3] that constrains each update with a **clipped surrogate objective**, keeping the new policy close to the old one for stable, sample-efficient training – the classical engine behind RLHF fine-tuning.



Direct Preference Optimization (DPO)

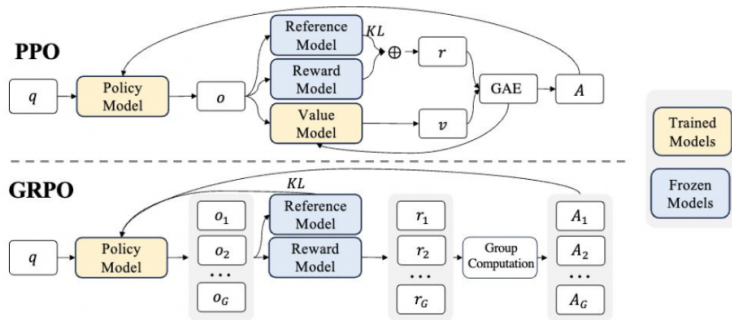
Reparameterizes the RLHF objective into a **closed-form classification loss** directly on preference pairs [4], showing that the **language model is implicitly its own reward model** – no explicit reward model or RL loop required.

- **Standard RLHF:** preferences $\rightarrow$ reward model $\rightarrow$ PPO fine-tuning.
- **DPO:** preferences $\rightarrow$ direct classification loss on the policy – simpler and more stable.



Group Relative Policy Optimization (GRPO)

A **PPO variant** that removes the learned value network, instead estimating the **advantage baseline from a group of sampled outputs** for the same prompt [5], reducing memory and compute – introduced to improve mathematical reasoning via RL.



Multi-Objective Alignment

Aligning an LLM against **several, potentially competing** human values (helpfulness, harmlessness, honesty) rather than a **single scalarized reward**, using **Pareto-aware or constrained** formulations of PPO/DPO/GRPO.

- **Scalarization:** weighted sum of objectives – simple but hides trade-offs.
- **Pareto-front methods:** expose the trade-off surface between competing objectives.
- **Constrained optimization:** optimize one objective subject to guarantees on the others.



Statistical Foundations of Preference Optimization

Theoretical questions of **sample complexity**, **convergence**, and **uncertainty quantification** for preference-based RL, building on **DPO**'s reduction of alignment to a **classification problem** [4].

- How much preference data is needed for a reliable policy update?
- Do PPO/GRPO-style updates converge under noisy preferences, and how fast?
- How to quantify uncertainty in the learned reward / policy?



RL/RLHF Libraries: TRL / trlX / OpenRLHF

Candidate **open-source frameworks** for prototyping the proposed algorithms.

- **TRL** (Hugging Face): PPO/DPO/GRPO trainers integrated with the transformers ecosystem.
- **trlX**: distributed RLHF training for large-scale models.
- **OpenRLHF**: efficient, large-batch RLHF training with support for multiple RL algorithms.



Benchmark Datasets & Test Environment

Resources used to **train** and **evaluate** the proposed robust, multi-objective RLHF framework.

- **Preference datasets:** Anthropic HH-RLHF, UltraFeedback.
- **Base models:** LLaMA-2/3, Mistral, GPT-2 (open-source).
- **Compute environment:** GPU cluster / HPC (ERIC Lab, Lyon 2).



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